

User Acceptance Testing (UAT)

Date	13-07-2026
Team ID	
Project Name	Nutrition Assistant – Nutrition Tracking Web Application (MERN)
Maximum Marks	

Project Overview

Project Name: Nutrition Assistant

Project Description: A MERN-stack based nutrition tracking web application that lets users search food items, view nutritional information, log daily calorie intake, monitor macronutrient consumption (protein, carbohydrates, fat), and track overall progress through an interactive dashboard.

Project Version: 1.0

Testing Period: [12-07-2026] to [13-07-2027]

Testing Scope

Features and functionalities tested:

- User registration and secure login/authentication (JWT-based).
- Personalized user dashboard with daily nutrition summary.
- Food search with nutritional information lookup.
- Daily calorie intake tracking and logging.
- Macronutrient analysis (protein, carbohydrates, fat).
- Data persistence and retrieval via MongoDB Atlas.
- Route protection / authorization for authenticated pages.
- Logout and session management.

User stories / requirements tested:

- As a user, I want to register and log in securely so that I can access my personal nutrition dashboard.
- As a user, I want to search for food items so that I can view their nutritional values.
- As a user, I want to log my daily food intake so that I can track my calorie consumption.
- As a user, I want to view my macronutrient breakdown so that I can maintain a balanced diet.
- As a user, I want to view a dashboard summary so that I can monitor my nutrition progress over time.

Testing Environment

URL/Location: <https://nutrition-assistant-ten.vercel.app> (Frontend) with backend API hosted on Render.

Credentials (if required): Test user account created via Registration page.

Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User registration with valid details	1. Navigate to Register page. 2. Enter valid name, email, and password. 3. Click Register.	Account is created successfully; confirmation shown and user redirected to Login page.	Account created and redirected to Login as expected.	Pass
TC-002	User registration with existing email	1. Navigate to Register page. 2. Enter an email already registered. 3. Click Register.	System should reject registration and show 'Email already exists' error.	Error message displayed; registration blocked.	Pass
TC-003	Registration with weak password	1. Navigate to Register page. 2. Enter a password below the minimum length/strength. 3. Click Register.	System should reject the password and display a validation error.	Validation error shown; registration blocked.	Pass
TC-004	Login with valid credentials	1. Navigate to Login page. 2. Enter registered email and correct password. 3. Click Login.	User is authenticated and redirected to the personalized dashboard.	Redirected to dashboard with nutrition summary displayed.	Pass
TC-005	Login with invalid credentials	1. Navigate to Login page. 2. Enter incorrect password. 3. Click Login.	System should deny access and show an appropriate error message.	Error message shown; access denied.	Pass
TC-006	Search for a food item	1. Log in to dashboard. 2. Navigate to Food Search. 3. Enter a food name and search.	Matching food items are displayed with basic nutritional details.	Food list loaded with matching results and nutrition info.	Pass
TC-007	View detailed nutrition info of a food item	1. From search results, select a food item.	Detailed nutrition values (calories, protein, carbs, fat) for the item are displayed.	Nutrition details displayed correctly for selected item.	Pass
TC-008	Log a food item to daily intake	1. Select a food item. 2. Enter quantity/serving size. 3. Click Add/Log.	Food item is added to the day's log; calorie and macro totals	Item logged successfully; totals updated instantly.	Pass

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
			update accordingly.		
TC-009	Attempt to log food with invalid quantity	1. Select a food item. 2. Enter zero or a negative quantity. 3. Click Add/Log.	System should reject the entry and show a validation error.	Error displayed; entry blocked.	Pass
TC-010	View daily calorie tracking	1. Log in. 2. Navigate to Calorie Tracker / Dashboard.	Total calories consumed for the day are displayed accurately based on logged foods.	Calorie total matched expected calculation.	Pass
TC-011	View macronutrient breakdown	1. Log in. 2. Navigate to Dashboard.	Protein, carbohydrate, and fat totals are displayed, matching logged food items.	Macronutrient breakdown displayed correctly.	Pass
TC-012	View nutrition summary on dashboard	1. Log in. 2. Open Dashboard page.	A summary of calories, macros, and overall progress is displayed for the current day.	Dashboard summary rendered with correct data.	Pass
TC-013	Access protected route without login	1. Log out or clear session. 2. Attempt to directly access a protected URL (e.g., /dashboard).	User should be redirected to the Login page and denied access.	Redirected to Login page; access denied as expected.	Pass
TC-014	Logout functionality	1. Log in. 2. Click Logout.	Session should end and user is redirected to the Login/Landing page.	Session ended and redirected correctly.	Pass
TC-015	Responsive UI on mobile viewport	1. Open the application on a mobile-sized viewport or device. 2. Navigate through Login, Search, and Dashboard.	Layout should adapt responsively with no broken elements or overlapping content.	UI rendered correctly across tested screen sizes.	Pass

Bug Tracking

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	Calorie total did not refresh immediately after logging a food item on slower network conditions.	1. Log in. 2. Add a food item. 3. Observe dashboard calorie total.	Low	Closed	Fixed by re-fetching dashboard totals immediately after a successful log action.
BG-002	Decimal precision inconsistency in displayed macronutrient values on the Dashboard page.	1. Log a food item with fractional gram values. 2. Compare displayed macro values with source data.	Medium	Closed	Standardized macro value formatting to one decimal place across the app.
BG-003	Food search allowed submission with an empty search field under rapid double-click conditions.	1. Open Food Search. 2. Rapidly double-click Search with the input field empty.	Medium	Closed	Added client and server-side validation to reject empty search queries.
BG-004	Dashboard macro chart did not resize correctly when switching between mobile and desktop viewports.	1. Open Dashboard on desktop. 2. Resize browser window to mobile width without reloading.	Low	Open	Scheduled for next sprint; workaround is to refresh the page after resizing.

Sign-off

Tester Name: M H S Rami Reddy

Date: 13-07-2026

Signature: M H S Rami Reddy

Notes

- All test cases above cover both positive scenarios (successful registration, valid food logging) and negative scenarios (invalid login, weak password, invalid quantity).
- Detailed feedback and suggestions for improvement are captured in the Bug Tracking section and Additional Feedback column.
- Bug tracking includes severity, current status, and steps to reproduce for each reported issue.
- Sign-off from both the project manager and product owner should be obtained before proceeding with deployment.